Felix Pernegger

01.09.2004
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EDUCATION

2025 – 2027 **University of Bonn** — Master of Science in Mathematics
- Current GPA 1.1/1.0
- Currently student research assistant
- Elected as student represantative to board of mathematical institute

2023 – 2025 **University of Bonn** — Bachelor of Science in Mathematics
- Final grade 1.0/1.0
- Honors program at University of Bonn
- Graduated one year early
- Bachelor thesis on Lean formalization of results in Ergodic theory (grade 1.0)

2015 – 2023 **B(R)G Ried im Innkreis** — Austrian High School Diploma (Matura)
- Pre-scientific thesis on Non-Euclidean geometry
- Graduated with distinction, achieving a final grade of 1.1/1.0

EXPERIENCE

Oct 2024 – Sep 2025 **Tutor**, University of Bonn
- Covering introductory topics in applied mathematics and introductory topics in (general / algebraic) topology
- Held weekly exercise sessions and graded homework and exams for 1st and 4th semester students
- Additionally voluntary Tutor for "Bonner Matheclub" since April 2025, holding presentations about interesting mathematical topics for talented school students.

Nov 2025 – Present **Student Research Assistant**, Bonn Formalisation Group
- Contributing to Lean's `mathlib` to formally verify mathematical results
- Working on a large formalisation project about convergence rates of multilinear ergodic averages

Nov 2025 – May 2026 **Consultant**, Project Numina
- Work on benchmark for Artificial Intelligence for mathematics
- Provide blueprints and formalisations in Lean to benchmark and train A.I. agents

AWARDS

2022	International Mathematical Olympiad (IMO): Honorable Mention
2021–2023	Various awards at Austrian Mathematical Olympiad, including first place nationally at contests in 2022 and 2023
2023	Advanced to the national stage of the Austrian Informatics Olympiad

PROJECTS

Euclidean Geometry Formalisation	Formalised planar Euclidean geometry from scratch in Lean, including Ceva's theorem (100 theorems list), approximately 14,000 lines of code.
Calderón Transference	Formalised a version of the Calderón Transference Principle from ergodic theory and harmonic analysis as part of bachelor thesis, approximately 9,000 lines of code.
Community Database of Topology	Contributor to the community database π-*base* in general topology, covering traits, theorems, and counterexamples; also working on an extensive formalisation thereof.

SKILLS

Languages	German (Native), English (C2), French (B1)
Programming Languages	Lean, Python